
4, 3, 6, 5

4: L given by $r(t) = \langle -2t, 2+7t, -1-4t \rangle$
intersects plane $4x+9y-2z+8=0$

Recall: 2 things intersect if the equations which describe them have simultaneous solutions.

So let's check for solutions by plugging $x=-2t$, $y=2+7t$, $z=-1-4t$ into plane equation:

$$\begin{aligned} 0 &= 8 + 4(-2t) + 9(2+7t) - 2(-1-4t) \\ &= 8 - 8t + 18 + 63t + 2 + 8t \\ &= 28 + 63t \end{aligned}$$

$$\Rightarrow \text{solution } t = -\frac{28}{63} = -\frac{4}{9}$$

So there will be an intersection, and you can find it by substituting $t = -\frac{4}{9}$ back in.

3: Plane through $P = (10, -1, 5)$ containing the line $\frac{x}{4} = y+6 = \frac{z}{5}$



This is not a convenient scenario, so let's try to translate it to a different scenario.

Let L be the line, A some point on the line, and $\vec{v}$ a direction vector for L . Notice that the plane contains A , $\vec{v}$ & $\vec{AP}$.

Note that any vector inside a plane is perpendicular to the normal of a plane, so if $\vec{n}$ is a normal, then $\vec{n} \perp \vec{v}, \vec{AP}$.

So we can take $\vec{n} = \vec{v} \times \vec{AP}$, and point + normal is enough to get the equation of a plane.

First, we need to find A & $\vec{v}$.

Plug in $y = -6$, then $\frac{x}{4} = 0 = \frac{z}{5}$

$\Rightarrow x = z = 0$, so $(0, -6, 0) \in L$,

so we can take $A = (0, -6, 0)$.

Notice the symmetric equations for L

are $\frac{x}{4} = \frac{y+6}{1} = \frac{z}{5}$, so

$\vec{v} = (4, 1, 5)$ is the direction of L
from last week's content.

$$\text{Next, } \vec{AP} = P - A = \langle 10, -1, 5 \rangle - \langle 0, -6, 0 \rangle \\ = \langle 10, 5, 5 \rangle,$$

$$\text{So } \vec{n} = \langle 0, -6, 0 \rangle \times \langle 10, 5, 5 \rangle \\ = -30 \langle 0, 1, 0 \rangle + \langle 2, 1, 1 \rangle$$

$$= -30 \langle 1, 0, -2 \rangle,$$

$$\begin{array}{l} 0 \ 1 \ 0 \\ 2 \ 1 \ 1 \end{array}$$

$$\therefore 1 \cdot 1 - 1 \cdot 0 = 1$$

$$\therefore -(0 \cdot 1 - 0 \cdot 2) = 0$$

$$\therefore 0 \cdot 1 - 1 \cdot 2 = -2$$

and $\langle 1, 0, -2 \rangle$ is also
normal for the plane

Since scaling doesn't affect normality.

So $\langle 1, 0, -2 \rangle$ normal & $(0, -6, 0)$ on plane

$$\Rightarrow 0 = 1(x-0) + 0(y+6) - 2(z-0)$$

$$= x - 2z \Rightarrow \boxed{x = 2z}$$

6: Domain $r = \langle \ln(t+1), \frac{t}{4-t^2}, \sqrt{4-t} \rangle$.

Let D_1, D_2, D_3 be domain of 1st-3rd
component. Recall a vector is defined
if and only if every component is.

D_1 : $\ln(-)$ is defined when argument > 0 ,

$$\text{so } t+1 > 0 \Rightarrow t > -1$$

D_2 : $t \in \mathbb{R}, 4-t^2$ always defined, so

$\frac{t}{4-t^2}$ defined unless $4-t^2=0 \Rightarrow t=\pm 2$

$D_3: \sqrt{\quad}$ is defined when $\text{arg} \geq 0$,

so $4-t \geq 0 \Rightarrow t \leq 4$

So you have to go from $-1 \rightarrow 4$

and skip $+2$, so domain of $r(t)$

is $\boxed{(-1, 2) \cup (2, 4]}$.

5: Which equation represents surface?

A is the equation of a sphere, and the graphs don't show a sphere, so it's not A.

Notice the 2nd graph shows the xz -plane cross section, which is when $y=0$.

It looks something like $z = cx^2$

for some $c > 0$.

Let's compare this to B, C, D when $y=0$.

B: $y=0 \Rightarrow z = 1-x^2 \Rightarrow$ inverted parabola

C: $z^2 = x^2 \Rightarrow z = \pm x \Rightarrow$ 2 lines

D: $z = 2x^2 \Rightarrow$ upward parabola

So $\boxed{\text{ans} = D}$ since it's the only one

that fits the 2nd graph

